Beautified Is Not Forged: A Two-Direction Measurement of Face-Forgery Detectors under Skin Smoothing

Anonymous submission

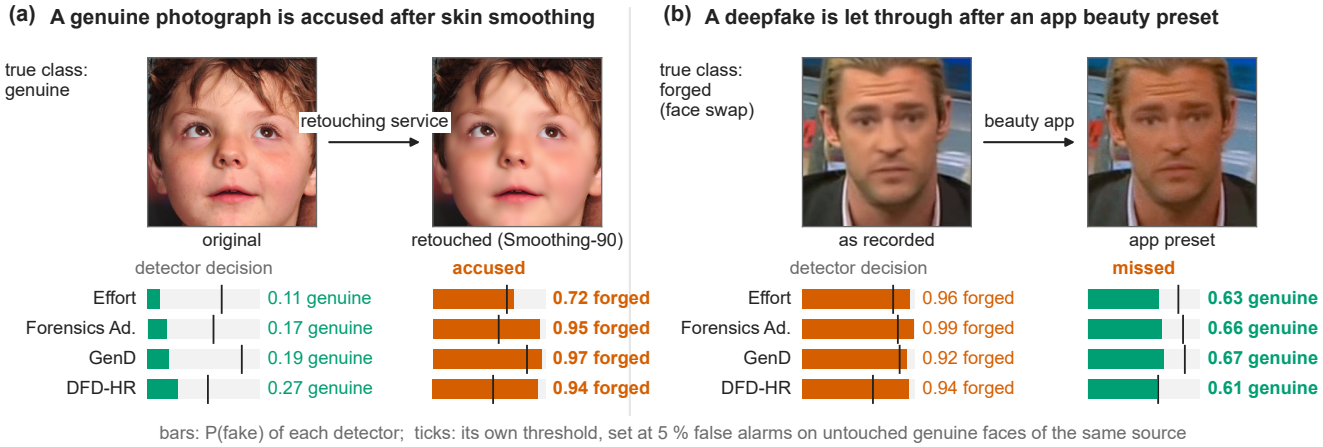


Fig. 1. One cue, two errors. (a) A genuine FFHQ photograph and its render by a commercial retouching service (skin smoothing, strongest level): the four CLIP-based detectors call the original genuine and the retouched copy forged. (b) A Celeb-DF-B face-swap video as recorded and after an app beauty preset: the same detectors catch the deepfake as recorded and miss it after the preset. Bars are $P(\text{fake})$, ticks are each detector's own threshold, set where it accuses 5% of the untouched genuine faces of the same source; (b) uses the video mean over 32 frames. Across seventeen published detectors the pattern is the rule, not the exception (Fig. 3).

***Abstract*—Published face-forgery detectors read skin smoothing as forgery evidence, and the error runs in both directions: a genuine face smoothed by a retouching service is accused, and a deepfake beautified by an app preset is let through. We score seventeen published detectors on pixel-aligned retouching pairs and on app-beautified deepfake videos with one decision rule. Fifteen of the seventeen score the smoothed genuine face above its own original in 89–99% of pairs, and fourteen miss 3–22 points more of the beautified deepfakes than of the untreated ones once re-encoding is controlled for. Controlled training arms show that the class that receives smoothed views during training sets the sign of the error, and that exposure to smoothing is necessary and sufficient for it. The cue is a face smoother than its own context, not blur and not the boundary line. Four repairs, from a conditional threshold to a consistency loss, each reduce to moving the operating point, because under smoothing the scores of genuine and forged faces coincide. The one channel we tested that beautification leaves intact is identity against a reference of the claimed person: on app-beautified face swaps it misses 0.6% of the forgeries at the false-alarm rate where texture detectors miss 46–72%, but its threshold still has to know about beautification. Manifests, scorer, paired reader, result files and the pre-registration log are released.**

I. Introduction

A person takes a selfie, lets a beauty app or a retouching service smooth the skin, and uploads it. The face is the same person with the same expression at the same moment; only the skin texture has changed. If that photograph reaches a forgery detector during content moderation, a remote identity check or fact-checking, it should be called genuine. If a deepfake of the same person passes through the same app, it should still be called forged. Beautification edits the pixels a detector inspects but changes neither who is shown nor what the person does. Skin smoothing, whitening, eye enlarging and face slimming are offered by camera apps, AR filters and retouching services [1]–[3]; we make no claim about how often they are used. Fig. 1 shows what happens today on one photograph and one video; across the seventeen published detectors we scored, the retouched copy outranks the original in 89–99% of pairs for fifteen of them (Fig. 3).

Three evaluations have looked at this. Libourel et al. [3] released Celeb-DF-B and report video-level AUC drops for three detectors under four app presets, without separating the two classes of error. Saeed et al. [4] apply six enhancement filters to deepfakes and measure how many escape six detectors; genuine faces are never enhanced, so the accusation direction is not measured. Concas et al. [5] apply one smoothing filter to genuine and forged test images of two CNNs they trained (AlexNet, VGG19) and observe that the two move in opposite directions, which they attribute to the detector ("the underlying system") without tracing it to training. Each study measures

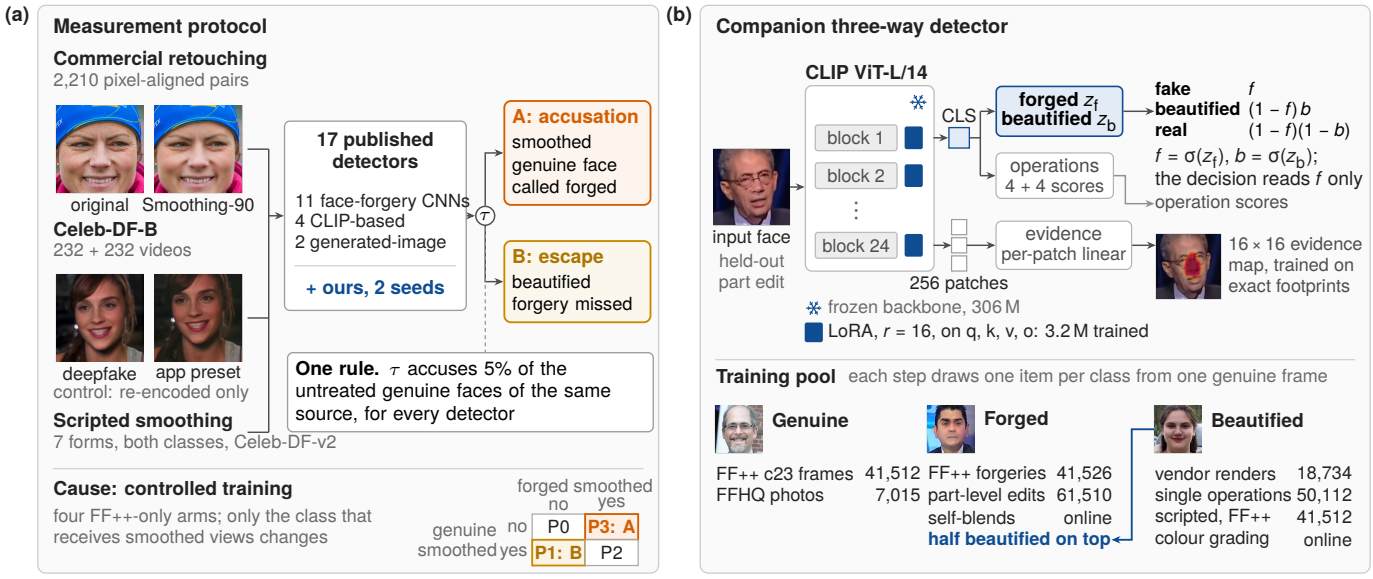


Fig. 2. Overview. (a) The measurement: beautified genuine and forged faces from three sources are scored by seventeen published detectors and ours, each thresholded at the score τ that accuses 5% of the untreated genuine faces of the same source; Direction A counts smoothed genuine faces called forged, Direction B beautified forgeries missed. The cause is tested with four training arms that differ only in which class receives smoothed views (Sec. III-C). (b) The companion detector of Sec. IV: a frozen CLIP ViT-L/14 with rank-16 LoRA, a forged and a beautified logit on the CLS token (the decision reads only f), auxiliary operation scores and a 16×16 evidence map on the patch tokens (shown on a held-out part-level forgery). Each training step builds one genuine, one forged and one beautified item from the same genuine frame; half of the forged items are beautified as well, so both classes see beautification.

one direction with a handful of detectors, and none tests a cause. In generated-image detection, Ricker et al. [6] showed that training with JPEG on one class only flips the sign of the error, with two of the four exposure cells. The implicit assumption shared by robustness benchmarks [7]–[9] is that a benign edit degrades a detector symmetrically, so that AUC under the edit is the number to report. We show that for skin smoothing the degradation has a sign, that the sign is set in training, and that AUC hides it.

We ask three questions. RQ1: across the detector families in use today, how often is a smoothed genuine face accused and how often does a smoothed forgery escape, under one rule? RQ2: where does the cue come from, which part of the edit carries it, and does it depend on what the detector was trained on? RQ3: can a decision rule or a training rule remove it without giving up detection of untreated forgeries?

Answering them needs four things. Both directions must be read on the same detector with the same rule; pairs must be read without a threshold where the original exists; re-encoding must be controlled and the cause tested by intervention on training; and uncertainty must be reported at the sampled unit, pairs or videos rather than frames.

We build such a measurement from three beautification sources in the main text: commercial retouching pairs, app-beautified deepfake videos and scripted smoothing (colour-filter and AR sources are in the supplement). We replicate the accusation on an open-source beauty engine and score seventeen published detectors and a family of our own controlled arms with one rule (Fig. 2). Our contributions are:

- A two-direction evaluation protocol and its results on seventeen detectors (Sec. II, Findings 1–2): on pixel-aligned commercial retouching pairs fifteen detectors rank the retouched copy above its original in nearly every pair, and on app-beautified deepfake videos fourteen miss more forgeries once re-encoding is removed; the skin smoothing operation by itself produces the shift.
- The cause by intervention (Findings 3–4). A complete 2×2 training-exposure design on face forgery, extending Ricker et al. [6] to all four cells, shows that the class receiving smoothed views sets the sign of the error, that removing smoothing from the views brings the error back and that smoothing alone sets it. A band-energy analysis and an edge-treatment ablation then show that the cue is the contrast between a flattened face and its untouched surroundings, neither texture loss nor the boundary line.
- Negative results that bound what a fix can do (Findings 5–6): no threshold rule separates smoothed genuine from smoothed forged faces for any of the detectors, and four repairs (a conditional threshold, a routed expert, a consistency loss, test-time context matching) each turn out to move the operating point rather than restore separation, since every model loses separability when both classes are smoothed.
- The channel that survives (Finding 7): identity against a reference of the claimed person keeps its separation under real app presets but not its operating point; the channel is prior art and we claim no fusion with it.
- An audit reference (Sec. IV): a three-way detector that sees smoothing on both classes, with a forgery-head-only control on the same data that does as well; it is trained on

more data than the published detectors and is not ranked against them.

II. Measurement Protocol

Classes. A face is *forged* if it shows a person, an expression, a motion or speech that was not captured: another identity in whole or in part, a reenactment, or a facial feature replaced by new content. It is *beautified* if its colour, tone, skin texture or the shape of the person's own features were edited after capture while identity, expression and pose stay those of the capture. The label follows what changed, not the tool, and a beautified forgery is forged. Cases near the boundary are listed in the supplement (Table S1); the boundary is stated, not measured.

One rule, both directions. Every detector is scored by its own $P(\text{fake})$ with the threshold that accuses 5% of each source's untreated genuine faces. Under beautification we report the false-alarm rate FA on beautified genuine faces (Direction A, the accusation) and the miss rate MISS on beautified forgeries (Direction B, the escape); a forgery that a three-way model sends to its beautified class counts as a miss. Where the untouched original of each beautified face exists we also report the share of pairs in which the beautified face scores above its own original, which needs no threshold and is the quantity of Fig. 3. Intervals are 95% percentile bootstraps over pairs or videos with the threshold re-estimated in every draw (1,000 draws), and differences between models use identical draws; a second bootstrap with identities as the unit changes none of the headline intervals (Table S14).

Sources. *Commercial retouching*: 2,210 FFHQ photographs [10] and their renders by one retouching service from RetouchingFFHQ [2], pixel-aligned, four operations at three levels; the strongest skin smoothing (Smoothing-90) is the headline condition, and the other operations serve as controls that pass through the same service and encoder. No model was trained on this service, but our companion model saw renders of the two other services of the same release (Sec. IV), and the service was read during the study, so it is an analysis set rather than a blind test.

App-beautified deepfake videos: Celeb-DF-B [3], 232 genuine and 232 forged Celeb-DF-v2 videos per condition (as recorded, re-encoded, beautified by one of four Instagram presets), 32 frames each; the rise in misses under a preset is read minus the rise under re-encoding, so what remains is the beautification alone.

Scripted smoothing: 4,016 Celeb-DF-v2 test frames (1,348 genuine, 513 videos) under seven forms applied to both classes: a whole-crop bilateral filter at three strengths, a 5×5 median, a 50% Gaussian blend, and a bilateral filter confined to the landmark face hull at two strengths (*face-local*, background untouched). *Calibration renders*: FF++ validation genuine frames under the same forms, used only to fit the threshold rule of Finding 5. *Untreated detection*: Celeb-DF-v2 [11], DFD [12] and DF40 [13]. Counts, crops and preprocessing are in the supplement.

What is not a clean source. Instagram AR filters rendered through OpenFilter [1] (FairBeauty, B-LFW) were scored but are excluded as evidence, for one reason: the filter cannot be separated from the capture pipeline. The AR engine re-renders the face through its own capture path, and there is no no-filter pass through that path. A model trained on such pairs still calls 94.8% of filtered images beautified when the face is greyed out, and a multi-operation arm whose AR accusation did not fall calls only 18.2% of the FairFace originals genuine (Sec. S6). Every detector, ours included, accuses 10–97% of these renders (Table S9); we report that as a limitation and make no claim on AR filters.

Detectors. Eleven face-forgery CNNs (Xception [14], EfficientNet-B4 [15], F3Net [16], SPSL [17], UCF [18], RECCE [19], CORE [20], SRM [21], SBI [22], CADDM [23], LAA-Net [24]), four CLIP ViT-L/14 detectors (Effort [25], Forensics Adapter [26], GenD [27], DFD-HR [28]) and two generated-image detectors (UnivFD [29], NPR [30]), all with released weights, each on its own crop. Table S3 sets the released checkpoints against their published Celeb-DF-v2 numbers: the eight DeepfakeBench CNNs land within 0.035 frame AUC of the published value, Forensics Adapter and GenD reproduce their video AUC, and Effort and DFD-HR fall 0.02–0.03 short and are used as released (Sec. V). CADDM and LAA-Net reproduce their video AUC within 0.011 on their own crops (Sec. S3); SBI, UnivFD and NPR have no comparable published number on this protocol. NPR saturates at 1.0 on the retouching source and is reported as such. Table S4 compares this protocol with the prior evaluations of Sec. I feature by feature.

III. Findings

A. Direction A: the accusation

Table I reads each Smoothing-90 render against its own original. Fifteen of the seventeen published detectors score the render above the original in 89–99% of pairs, and the median $P(\text{fake})$ moves from 0.21 to 0.99 for SBI and from 0.23 to 0.92 for GenD (Table S20): a heavily smoothed genuine face is scored like a forgery whatever the threshold. The two exceptions are NPR, whose scores saturate, and CADDM, which reads a boundary rather than texture and shifts in 60.9% of pairs. At the threshold that accuses 5% of the originals, the published detectors accuse 17–69% of the renders (CADDM 6.3%, NPR 0%). Only smoothing produces the shift: whitening, face slimming and eye enlarging at their strongest level stay at 3.7–8.1%, close to the 5% floor, for thirteen of the sixteen detectors other than NPR, although they pass through the same service, encoder and resolution.

Skin-only smoothing. A bilateral filter confined to the face hull, with the background untouched, is accused by all seventeen detectors (Table S12): the four CLIP detectors call 88–99% of genuine frames forged at the milder strength and 99.9–100% at the stronger one, the eleven CNNs 64–100% and 92–100%, UnivFD 14 and 55%. For the four CLIP detectors the forgery logit rises in at least 97.9% of genuine frames at every strength, the weakest face-local level by 1.0–1.9 times the gap between the mean forged and mean genuine logit (Table S13).

TABLE I

DIRECTION A ON 2,210 PIXEL-ALIGNED RETOUCHING PAIRS. ABOVE OWN: SHARE OF PAIRS IN WHICH THE SMOOTHING-90 RENDER SCORES ABOVE ITS OWN ORIGINAL (50 MEANS NO SHIFT). FA: SHARE OF RENDERS ABOVE THE THRESHOLD THAT ACCUSES 5% OF THE ORIGINALS. RIGHT: THE SAME FA FOR WHITENING, FACE SLIMMING AND EYE ENLARGING AT THEIR STRONGEST LEVEL. BRACKETS: 95% BOOTSTRAP OVER PAIRS. NPR SCORES EVERYTHING AT 1.0; –: NOT SCORED. BOLD: ABOVE OWN CLOSEST TO 50 AND THE LOWEST SMOOTHING-90 FA (NPR NOT RANKED). OURS (BOLD): THE THREE-WAY DETECTOR OF SEC. IV, TWO SEEDS; IT WAS TRAINED ON RENDERS OF TWO OTHER SERVICES FROM THE SAME RETOUCHINGFFHQ RELEASE AND ON MORE DATA THAN THE PUBLISHED DETECTORS, SO ITS ROWS ARE A REFERENCE, NOT A RANKING.

	Smoothing-90		FA, other operations (level 90)		
Detector	above own [CI]	FA [CI]	white	slim	eye
Xception	95.2 [94.3, 96.1]	51.6 [49.5, 53.7]	5.7	3.7	4.6
EfficientNet-B4	93.7 [92.6, 94.7]	36.3 [34.3, 38.4]	5.1	4.0	4.7
F3Net	97.8 [97.1, 98.3]	50.6 [48.5, 52.6]	5.4	4.2	4.4
SPSL	97.3 [96.7, 98.0]	51.2 [49.1, 53.3]	5.2	4.9	4.9
UCF	95.4 [94.5, 96.2]	41.4 [39.4, 43.3]	6.6	4.0	4.2
RECCE	96.5 [95.7, 97.2]	36.8 [34.8, 38.8]	5.3	4.1	4.5
CORE	96.6 [95.8, 97.3]	54.8 [52.7, 56.8]	5.7	4.4	4.8
SRM	99.3 [98.9, 99.6]	62.8 [60.8, 64.8]	4.5	3.8	4.7
SBI	96.2 [95.4, 97.0]	37.3 [35.2, 39.2]	5.4	6.2	5.5
CADDM	60.9 [58.8, 62.9]	**6.3** [5.3, 7.4]	4.7	6.0	5.9
LAA-Net	89.2 [87.8, 90.5]	17.2 [15.7, 18.9]	5.2	4.8	4.8
UnivFD	90.4 [89.1, 91.7]	16.6 [15.0, 18.1]	6.2	5.7	11.7
NPR	31.8 [29.7, 33.6]	0.0 [0.0, 0.0]	n/a	n/a	n/a
Effort	95.6 [94.8, 96.5]	43.2 [41.1, 45.3]	11.1	9.6	16.8
Forensics Adapter	93.3 [92.3, 94.4]	39.7 [37.8, 41.9]	5.4	6.2	7.0
GenD	98.5 [97.9, 99.0]	68.7 [66.8, 70.7]	8.1	6.4	7.5
DFD-HR	99.0 [98.6, 99.5]	64.9 [62.8, 66.8]	7.7	8.0	9.6
OURS, seed 1	32.5 [30.5, 34.4]	8.6 [7.4, 9.7]	4.8	3.7	3.6
OURS, seed 2	**41.3** [39.3, 43.3]	12.5 [11.1, 13.8]	–	–	–

Finding 1. *Every one of the seventeen published detectors we scored, across four families and seven years, scores a skin-smoothed genuine face as more forged than the same face untouched (NPR only on the scripted forms, where its scores do not saturate); smoothing the skin alone is enough, and the other beautification operations do not produce the shift.* The effect is not a threshold artefact: it appears in the pair ranking before any threshold is set. It holds for detectors trained on FF++ and for UnivFD, which was not, so it is not a property of one training set. Evaluations of benign processing should report the paired ranking before any AUC.

B. Direction B: the escape

Table II reads the same detectors on deepfake videos beautified by a real app. On this source the accusation is small, 2–15% of beautified genuine frames against a 5% floor for most detectors, and the escape is not. Once the rise under re-encoding is subtracted, fourteen of the seventeen detectors miss 3–22 points more of the beautified forgeries, with the interval above zero. The largest rises belong to SBI, SRM and Forensics Adapter. Every detector that detects the untreated forgeries loses 0.04–0.17 of frame AUC beyond re-encoding (LAA-Net 0.870 to 0.698, SBI 0.861 to 0.752). Libourel et al. [3] reported the AUC drop for three of these detectors; the one-rule reading adds that the drop is almost entirely misses.

Finding 2. *The same detectors that accuse a smoothed genuine photograph let an app-beautified deepfake video through: beautification moves forgeries toward the genuine side by 3–22 points for fourteen detectors, and the loss is paid in misses rather than false alarms.* The two directions appear on different sources because the operating point differs; Finding 5 shows that they are one shift seen from either side of the threshold. For the uploader of Sec. I this is a 17–69% chance of accusation at the strongest smoothing level, and for a remote identity check a 3–22 point higher chance that a beautified swap passes; both axes for every detector are in Fig. 5.

TABLE II

DIRECTION B ON CELEB-DF-B. AUC ON UNTREATED FRAMES AND ON FRAMES BEAUTIFIED BY THE APP PRESETS (FRAME LEVEL), FA ON BEAUTIFIED GENUINE FRAMES, AND THE RISE IN THE MISS RATE UNDER BEAUTIFICATION MINUS THE RISE UNDER RE-ENCODING ALONE, AT THE 5%-FPR THRESHOLD OF UNTREATED GENUINE FRAMES. BRACKETS: 95% BOOTSTRAP OVER VIDEOS. BOLD: HIGHEST AUC, LOWEST FA_b AND THE ΔMISS CLOSEST TO ZERO PER COLUMN (NPR, AT CHANCE, IS NOT RANKED).

Detector	AUC untr.	AUC beautified [CI]	FA_b	ΔMISS beauty only [CI]
Xception	0.811	0.737 [0.702, 0.773]	5.7	+10.9 [7.4, 14.1]
EfficientNet-B4	0.811	0.762 [0.729, 0.798]	7.8	+3.4 [0.9, 5.9]
F3Net	0.798	0.731 [0.697, 0.767]	5.4	+10.9 [7.4, 14.8]
SPSL	0.781	0.727 [0.690, 0.762]	3.4	+12.4 [9.3, 15.5]
UCF	0.841	0.777 [0.746, 0.808]	5.7	+11.3 [7.6, 15.4]
RECCE	0.785	0.723 [0.687, 0.758]	4.1	+13.8 [10.7, 17.1]
CORE	0.821	0.752 [0.718, 0.786]	4.0	+13.5 [10.2, 16.6]
SRM	0.789	0.733 [0.698, 0.768]	2.8	+15.3 [12.1, 18.7]
SBI	0.875	0.752 [0.718, 0.785]	**1.8**	+21.8 [17.3, 26.1]
CADDM	0.886	0.762 [0.730, 0.790]	13.8	**−2.9** [−5.8, 0.1]
LAA-Net	0.889	0.698 [0.660, 0.737]	6.0	+14.7 [10.9, 18.7]
UnivFD	0.653	0.624 [0.579, 0.663]	7.6	−7.2 [−10.2, −4.9]
NPR	0.503	0.498 [0.453, 0.543]	3.4	−0.7 [−2.2, 0.4]
Effort	0.909	0.807 [0.778, 0.836]	5.4	+11.0 [7.7, 14.5]
Forensics Adapter	0.909	0.828 [0.801, 0.856]	7.9	+15.2 [11.9, 18.6]
GenD	0.924	0.807 [0.776, 0.836]	9.9	+12.0 [8.2, 15.6]
DFD-HR	0.938	0.838 [0.807, 0.864]	14.6	+3.6 [0.6, 6.8]
OURS, seed 1	0.938	0.866 [0.841, 0.890]	9.5	+6.7 [4.0, 9.5]
OURS, seed 2	**0.941**	**0.883** [0.861, 0.902]	6.7	+8.3 [5.6, 11.0]

TABLE III

THE 2×2 EXPOSURE DESIGN. FF++-ONLY CLIP VIT-L/14 LORA ARMS, FORGERY HEAD ONLY, IDENTICAL EXCEPT FOR WHICH CLASS RECEIVES BEAUTIFIED TRAINING VIEWS (FACE-LOCAL SKIN SMOOTHING AMONG OTHER EDITS). SMOOTHING-90: RENDERS ACCUSED AT THE 5%-FPR THRESHOLD OF THEIR ORIGINALS [95% OVER ORIGINALS]. ΔMISS: CHANGE OF THE MISS RATE ON STRONGLY WHOLE-CROP-SMOOTHED CELEB-DF-V2 FORGERIES. CDF-B: BEAUTIFICATION-ONLY MISS RISE, THREE SEEDS. CDFV2: UNTREATED VIDEO AUC. P3 HAS ONE SEED.

Arm	smoothed views to	Smoothing-90 [CI]	ΔMISS	CDF-B ΔMISS, 3 seeds	CDFv2
P0	neither class	54.2 [50.5, 58.3]	−16.0	+17.0 / +14.2 / +10.8	0.938
P1	genuine only	3.7 [2.9, 4.4]	+51.2	+21.5 / +21.8 / +20.4	0.900
P2	both classes	11.7 [9.7, 14.3]	−26.5	+9.8 / +10.3 / +12.8	0.917
P3	forged only	63.8 [59.0, 67.8]	−38.5	–	0.943

C. The cause: training exposure

To find where the cue comes from we hold architecture, data and recipe fixed and vary only which class receives beautified views during training (Fig. 4, Table III). The arm without beautified views, P0, already accuses 54% of the Smoothing-90 renders, so an FF++-trained detector reads smoothing as forgery evidence out of the box. Giving the views to genuine faces only (P1) removes the accusation and opens an evasion route. On Celeb-DF-B its beautification-only miss rise exceeds P0's by 4.5–9.5 points in every one of three seeds (Holm-adjusted $p \leq 0.01$). On frames it has never seen it misses 96–

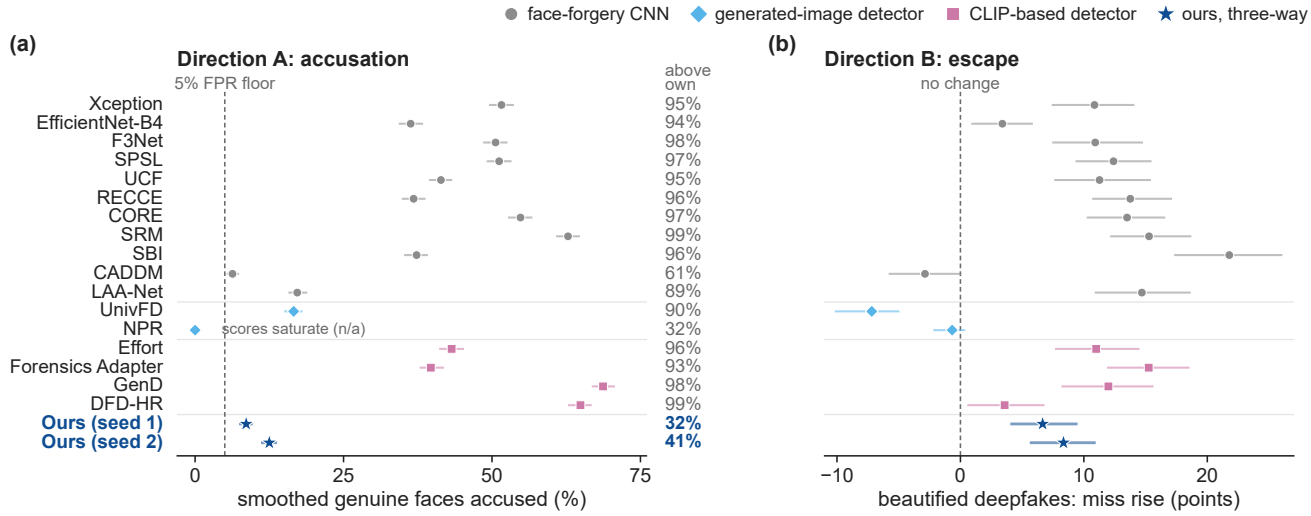


Fig. 3. The two directions on seventeen published detectors and ours, one row per detector. (a) Direction A: share of the 2,210 Smoothing-90 renders accused at the threshold that accuses 5% of the untouched originals (dot, 95% interval over pairs; dashed line: the 5% floor); the grey number is the threshold-free reading, the share of pairs in which the render scores above its own original. NPR is n/a because its scores saturate. (b) Direction B: rise of the miss rate on Celeb-DF-B deepfakes from the app presets alone, re-encoding subtracted (95% video-cluster intervals). Families from top to bottom: face-forgery CNNs, generated-image detectors, CLIP-based detectors, ours (two seeds).

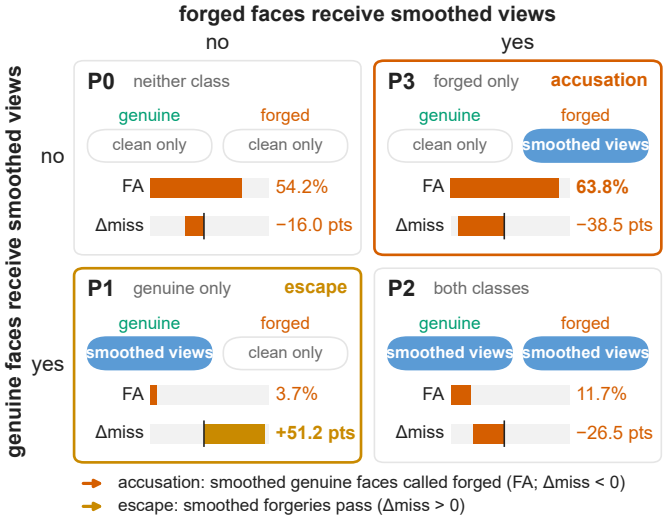


Fig. 4. The class that receives smoothed training views sets the direction of the error. Four FF++-only arms with the same architecture, data and recipe, first seed. FA: Smoothing-90 renders accused at the threshold that accuses 5% of their originals. Δmiss: change of the miss rate on Celeb-DF-v2 forgeries under strong whole-crop smoothing at the same rule; a negative change (dark orange) means the arm accuses every smoothed face, a positive one (amber) that smoothed forgeries pass. P1 opens an evasion route; P3 accuses.

100% of bilaterally smoothed forgeries when smoothing is the only view (a miss rise of 56–63 points on three seeds). Giving the views to forged faces only (P3) mirrors P1: it accuses 63.8% of the renders, 98–100% of smoothed genuine frames, and catches smoothed forgeries more often than untreated ones. Views on both classes (P2) sit in between, and on Celeb-DF-B the P2 minus P0 difference changes sign across seeds, so symmetric exposure is on average better than none but not reliably so (Table S15).

Two further arms show that smoothing is the operative view. Removing smoothing from the both-class views while keeping colour grades, filters, whitening and geometric edits brings the accusation back to 55–67% (exposure to smoothing is necessary), and smoothing as the only view on both classes lowers it to 2–5% (it is sufficient). The arms do not make the classes more separable; P3's untreated Celeb-DF-v2 video AUC is 0.943 against 0.938 for P0, and exposure moves errors between misses and false alarms.

Finding 3. *The class that receives smoothed views during training sets the sign of the error, and exposure to smoothing is both necessary and sufficient to set it: genuine-only views teach "smoothed means genuine" and open an evasion route, forged-only views teach the opposite, and no other beautified view substitutes.* This is shortcut learning [31] under class-dependent augmentation [32], which Ricker et al. [6] saw for JPEG in two of the four cells; here all four are measured on face forgery, three of them with replication. In practice this is a warning: a "filter-robust genuine class", the recipe a defender would reach for first, is the P1 cell.

D. The cue: a face smoother than its context

If the detectors reacted to lost texture, the forms that remove more high-frequency energy would be accused more. They are not (supplement, Fig. S1): across seven scripted forms and seventeen detectors the false-alarm rate does not follow the energy removed above 0.25 cycles per pixel (pooled Spearman +0.06); the Gaussian blend removes the most energy and is accused least, the face-local filter removes less and is accused by nearly every detector. Three contrasts show what the detectors do follow. With the same kernel, confining the smoothing to the face raises the accusation by 28 points (16

TABLE IV

A SMOOTHING-CONDITIONAL THRESHOLD. FIXED: FAKE IFF $s > t_0$ WITH t_0 AT 5% FPR ON UNTREATED GENUINE FRAMES. CONDITIONAL: $t_0 + \Delta(b)$ WITH Δ LEARNT ON FF++ VALIDATION GENUINE RENDERS ONLY, b THE SMOOTHING-PRESENCE SCORE OF OUR THREE-WAY MODEL. FA ON FACE-LOCAL SMOOTHED GENUINE FRAMES (STRENGTH 0.12) AND MISS ON THE SAME SMOOTHING APPLIED TO FORGERIES, FIXED → CONDITIONAL. AUC: BOTH CLASSES SMOOTHED, SAME FORM. RECAL. MISS: MISS RATE WHEN THE THRESHOLD IS RE-SET AT 5% FPR ON THE SMOOTHED GENUINE FRAMES THEMSELVES (AN ORACLE). CELEB-DF-V2 FRAMES WITH LANDMARKS.

	FA (%)		MISS (%)		AUC	Recal.
Detector	fixed	cond.	fixed	cond.	both sm.	MISS
Xception	99.3	92.4	0.2	4.2	0.580	90.7
SBI	100.0	0.0	0.0	100.0	0.577	93.4
CADDM	99.9	0.0	0.0	100.0	0.572	91.7
Effort	99.9	0.0	0.0	100.0	0.658	84.0
Forensics Adapter	100.0	0.0	0.0	100.0	0.649	87.6
GenD	100.0	11.7	0.0	78.3	0.632	90.8
DFD-HR	100.0	0.6	0.0	99.3	0.601	93.5
OURS, forgery head only	84.5	0.1	2.5	100.0	0.755	73.7

of 17 detectors, $p = 5 \times 10^{-4}$). At equal energy loss an edge-preserving filter is accused 67 points more than a uniform blur (16 of 17). Only the band 0.02–0.12 cycles per pixel, the skin relief that a bilateral filter flattens and a Gaussian blend leaves, correlates with the accusation ($\rho = +0.45$).

A pre-registered edge ablation locates the cue (Sec. S20). At the same strength, smoothing only a band at the face edge is accused at a median of 12% (7–51%) by the thirteen face detectors other than CADDM (94%) and LAA-Net (65%), which are trained on blending boundaries, against 92–100% for the whole face. Smoothing face and surroundings alike is accused much less (UCF 13.6%, Effort 30.3%). Nor do the texture-poor FF++ families teach it: an FF++-only arm trained without Deepfakes and NeuralTextures accuses face-local smoothing at 95.5 and 100% (P0: 95.8 and 100%; Sec. S22), and UnivFD, which never saw FF++, accuses it too.

Finding 4. *The shared cue is a face that is smoother than its own context: skin flattened with kept edges inside the face while the surroundings keep their texture. It is not blur, not texture loss as such and not the boundary line.* This is the face-versus-context inconsistency that face-swap detectors are built to find, which face-local beautification reproduces by construction. It is why Gaussian blur or JPEG as a training view does not remove the accusation (Finding 6; P0 itself trains with blur and JPEG, Sec. S17) and why a training-time fix has to use the smoothing signature itself.

E. A threshold cannot separate

A fixed threshold accuses smoothed genuine faces (Finding 1). The natural repair is to raise the threshold when smoothing is detected. We fit such a rule with the smoothing-presence output of our three-way model as the covariate and the shift learnt on FF++ validation genuine renders only, never on test data (Table IV; every detector in Table S28). It works on the genuine side: the accusation of strongly smoothed genuine faces falls to 0–1% for Effort, Forensics Adapter, DFD-HR, CADDM and our forgery head. It fails on the forged side at the same moment: 78–100% of the smoothed forgeries now pass. The reason is in the sixth column: with both classes smoothed, the AUC between smoothed genuine and smoothed forged faces is 0.60–0.66 for the four CLIP detectors and 0.57–0.58 for SBI and CADDM, so the two score distributions overlap and any threshold that clears one clears the other. Even an oracle that re-sets the threshold on the smoothed genuine frames themselves leaves every published detector missing 67–100% of the smoothed forgeries at the two strong forms, and ours 52–77% (Table S26).

Finding 5. *No threshold rule separates smoothed genuine from smoothed forged faces for any of the detectors: under skin smoothing the two score distributions coincide, so the accusation and the escape are one distribution read from two sides, and a decision rule can only move the error from one side to the other.* Calibration on untreated data, the remedy proposed for smartphone photographs [33], does not apply here, because the problem is not where the threshold sits but that there is no threshold to find.

F. Training only trades

If the cue is set in training, can training remove it? We tested the rule that Finding 3 and Finding 4 suggest: one skin-smoothing operator drawn identically for every training item of every class, and a penalty λ on the change of the forgery logit between the clean item and its smoothed copy, with the clean logit held as a stop-gradient target. Bars were written before training (Table S24). Table V gives the family and Fig. S3 in the supplement plots it. Every penalty weight from 0.03 upward removes the face-local accusation (3–7% on smoothed genuine frames, against 52 and 85% for the reference), and the symmetric view alone ($\lambda = 0$) does not (54 and 92% at the two face-local strengths), so the penalty does the work. The commercial renders do not separate the arms: the reference already accuses only 9.1% of them and the penalty arms 2–13%.

The cost has two faces that trade against each other along λ. At large weights the clean forgery score is pulled toward its smoothed copy and untreated detection falls (Celeb-DF-v2 video AUC 0.907 at $\lambda = 1$ against 0.956 without the rule); at small weights detection is kept (0.946 at $\lambda = 0.03$) but the smoothed forgery's score falls with the genuine one and its miss rises by 24 points. A one-sided penalty that forbids only a rise of the logit is the limit of that direction: untreated AUC 0.954 and 56 points more smoothed forgeries missed. A Gaussian blur of the same support in place of the smoothing operator fails the accusation bar at the strong face-local form (19.5%) and opens the bilateral evasion. DF40 reenactment is 0.06–0.09 below the reference at every weight, the one bar no member passes.

The penalty buys the operating point and leaves separability where the view put it. The deciding control is the $\lambda = 0$ model read with the oracle smoothing-conditional threshold of Finding 5. With both classes smoothed it keeps an AUC of 0.832 (bilateral) and 0.764 (face-local), against 0.790 and 0.778 for SIC at $\lambda = 1$ and 0.802 and 0.778 at $\lambda = 0.1$. Its

TABLE V

SMOOTHING-INVARIANT CONSISTENCY (SIC) AGAINST THE SAME MODEL TRAINED WITHOUT IT (FORGERY HEAD ONLY, SAME DATA, VIEWS, SEEDS AND CHECKPOINT RULE). FA: SMOOTHED GENUINE CELEB-DF-V2 FRAMES CALLED FAKE AT THE 5%-FPR RULE (FACE-LOCAL 0.06 / 0.12, WHOLE-CROP BILATERAL 15); S-90: COMMERCIAL SMOOTHING-90 RENDERS ACCUSED; ΔMISS: MISS RISE ON FACE-LOCAL 0.12 SMOOTHED FORGERIES; UNTREATED VIDEO AUC ON CELEB-DF-V2 AND DFD; CELEB-DF-B AUC UNDER THE PRESETS; DF40 REENACTMENT VIDEO AUC (FF++ DOMAIN); BARS PASSED OF SIX PRE-REGISTERED BARS. THE VIEW-ONLY CONTROL ($\lambda = 0$) DRAWS THE SAME SMOOTHED COPY FOR BOTH CLASSES WITHOUT THE PENALTY.

Model	seed	smoothed genuine accused (%)			S-90	Δmiss	untreated video AUC		Celeb-DF-B	DF40 FR	bars
		fl 0.06	fl 0.12	bil 15	acc. %	fl 0.12	CDFv2	DFD	AUC	ff	passed
forgery head only, no rule	1	51.6	84.5	35.5	9.1	−34.0	0.956	0.987	0.866	0.842	–
symmetric view only ($\lambda = 0$)	1	53.9	91.8	1.6	11.9	−40.1	0.943	0.975	0.891	0.835	3/6
SIC, $\lambda = 0.03$	1	5.0	3.3	9.6	11.2	24.4	0.946	0.979	0.886	0.750	4/6
SIC, $\lambda = 0.03$	2	3.8	3.5	8.8	15.0	20.4	0.933	0.975	0.876	0.777	2/6
SIC, $\lambda = 0.1$	1	6.5	4.2	9.9	10.5	15.8	0.930	0.980	0.871	0.774	3/6
SIC, $\lambda = 0.1$	2	4.6	4.6	8.5	8.1	17.6	0.929	0.974	0.868	0.771	2/6
SIC, $\lambda = 0.3$	1	6.0	4.2	8.1	12.8	9.5	0.898	0.971	0.840	0.692	2/6
SIC, $\lambda = 1$	1	5.8	4.4	5.8	2.1	9.7	0.907	0.986	0.835	0.768	2/6
SIC, $\lambda = 1$	2	6.2	5.5	6.3	12.3	6.1	0.844	0.968	0.804	0.746	2/6
one-sided penalty	1	0.1	0.1	0.0	9.2	56.5	0.954	0.980	0.880	0.794	4/6
one-sided penalty	2	0.1	0.0	0.0	10.4	47.3	0.929	0.975	0.860	0.775	2/6
Gaussian view (ablation)	1	5.8	19.5	1.1	3.5	−13.5	0.855	0.971	0.819	0.782	1/6
Gaussian view (ablation)	2	5.9	14.5	1.3	8.7	−9.3	0.848	0.960	0.821	0.784	2/6
SIC, $\lambda = 1$, FF++ only	1	7.4	6.8	5.3	7.0	4.8	0.856	0.979	0.829	0.674	2/6

recalibrated miss rate (57 and 77%) is not higher than SIC's (64–73% on two seeds). SIC therefore amounts to a symmetric view plus one threshold shift that carries across untreated and smoothed inputs; it does not keep separation that the view alone would lose.

Nor does a routed expert. A second model trained only on smoothed copies, selected by the smoothing score, clears the accusation (face-local 51–52 and 80–85% to 1–2%) and raises the miss on smoothed forgeries by 41–46 points; its both-classes-smoothed AUC is below the untreated detector's on seven of the eight forms, on both seeds (Table S27). Nor does a rule at test time. Calling a face forged only if it also exceeds the threshold after its whole crop is smoothed, which removes the face-versus-context contrast, clears the accusation for 10 of 15 face detectors. At a matched false-alarm rate on the smoothed genuine frames, however, it misses smoothed forgeries within a few points of the recalibrated single score for all fifteen (Sec. S20).

The loss is not confined to the shortcut arms: with both classes smoothed, every one of the seventeen published detectors loses AUC, by 0.06–0.28 at the strong face-local form for the sixteen that are above chance, and the $\lambda = 0$ model, which saw smoothed views of both classes in training, still loses 0.061 and 0.129. Part of the forgery evidence is erased by smoothing itself, beyond the shortcut; how much of the published detectors' loss comes from the shortcut being pushed to its ceiling and how much from erased evidence cannot be separated with these data, and we call the $\lambda = 0$ loss a measured loss, not a bound.

Finding 6. *Training can move the error but not remove it. A class-symmetric smoothing view with a consistency penalty takes the accusation of smoothed genuine faces from 85% to 4% at a cost in untreated detection, but it is a smoothing-conditional recalibration, since the view-only model read with a conditional threshold separates smoothed faces at least as well. A routed expert and test-time context matching are recalibrations too, and with both classes smoothed every detector, published or ours, loses separability.* A fix for the accusation has to be bought with misses on smoothed forgeries or with untreated detection, and a paper that reports only one of the three is reporting a point on this curve, not a solution.

G. Identity: the channel that survives

Smoothing erases texture evidence but not who the person is. We score the ArcFace distance between each Celeb-DF-B frame and a reference of the claimed person, made from up to four of that person's Celeb-DF-v2 genuine videos outside the benchmark, with four predictions written before any frame was embedded (Table S31; scripted forms and ICT-Ref [34] in Tables S29 and S30). Its video AUC is 1.000 untreated, beautified and re-encoded. Re-thresholded to accuse 5% of the beautified genuine frames, it misses 0.6% of the beautified forgeries, against 46–72% for the texture detectors (ours 51.1 and 46.2%, Effort 51.9%, SBI 69.6%, CADDM 72.1%). Its operating point still moves: at the untreated threshold its false alarm on beautified genuine frames rises by 8.5 points [6.4, 10.6] beyond re-encoding, short of the predicted 10. A rank-mean fusion with a texture detector meets the pre-declared bars with our model (false-alarm rise +7.7, miss rise +2.9) but not with Effort (miss rise +8.5 [5.4, 11.5] against a bar of 5), so no fusion method is claimed. Two of the four predictions hold (the stable AUC and the recalibrated miss) and two do not (the fusion bars and the size of the operating-point shift).

Finding 7. *A recognition embedding compared with a reference of the claimed person is the one channel we tested that beautification leaves: under real app presets on face swaps its separation does not move while every texture detector loses most of its own. Its operating point does move, so a detector that uses it needs a beautification-aware threshold.* The channel needs a reference, without which it is near chance

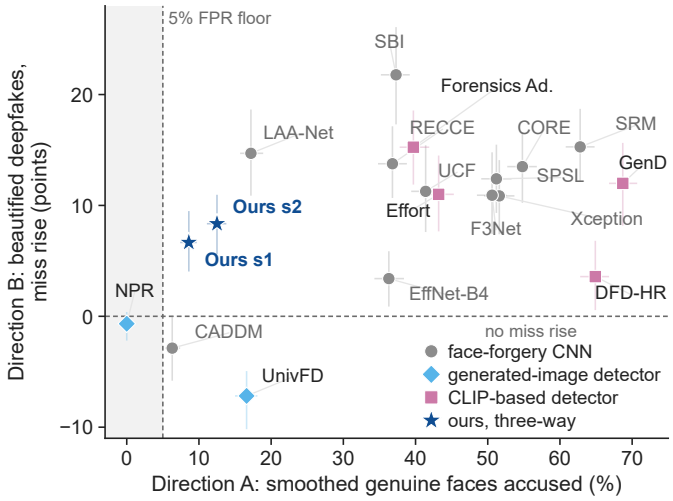


Fig. 5. No published detector is low on both axes; ours and CADDM sit in the lower-left corner. Each point is one detector: Direction A, Smoothing-90 renders accused at the threshold that accuses 5% of the originals (Table I), against Direction B, the beautification-only miss rise on Celeb-DF-B (Table II); bars are the 95% intervals, NPR is saturated at zero, EffNet-B4 is EfficientNet-B4. The lower-left corner holds our two seeds and CADDM, whose interval contains zero but which accuses 13.8% of beautified genuine frames.

(AUC 0.59); it is blind to reenactment and attribute edits, which keep the identity; and Celeb-DF swaps are easy for it, so the 1.000 is an upper bound. Identity evidence is prior art [34]–[37], including detectors personalised on reference images of a protected identity [38], and the stable-AUC, moving-threshold pattern under beauty filters is known from face recognition [39]; those works build detectors, whereas the measurement of the channel under beautification is new. A reference of the claimed person does not, however, catch the beautification itself. Across one person's untreated videos, the share of score variance that belongs to the person (intraclass correlation, ICC) is 0.14 for our beautified score, 0.12–0.46 for the texture detectors and 0.06–0.32 for skin texture and colour; subtracting the person's reference mean leaves the preset-only AUC where it was (ours 0.720 and 0.718, gain -0.002 $[-0.015, 0.011]$; seven texture detectors -0.023 to $+0.005$). Face width is identity-stable (ICC 0.66): on the benchmark's genuine videos narrowed by 2.25%, the same reference lifts the video AUC of a landmark width ratio from 0.635 to 0.727 (gain $+0.092$ [0.067, 0.116], short of the pre-declared $+0.10$), while our beautified score gains nothing on the same images. The variation that hides a light colour or texture edit belongs to the capture, not to the person, so a reference helps only for edits of the person's geometry, the part that differential retouching detection reads from landmarks [40]. A 3D morphable-model fit [41] is steadier across one person's videos but registers 0.2% of the 2.25% narrowing, so it is the wrong instrument for edits of this size.

IV. A Companion Three-Way Detector

Our companion model (Fig. 2b) is the detector that Findings 3 and 6 call for: smoothing on both classes in training, and the forgery decision read from a head that is not asked to be smoothing-invariant. Its beautified attribute is optional, since on the same data a forgery-head-only model is at least as good on every forgery number and better on reenactment, and the model was trained on more data than the published detectors, so the comparisons below are an audit reference, not a ranking.

System and training recipe. The backbone is a frozen CLIP ViT-L/14 [42] with rank-16 LoRA [43] on the query, key, value and output projections of all 24 blocks (3.2 M of 306 M parameters trained). Two linear heads on the CLS token give the forged logit z_f and the beautified logit z_b; with $f = \sigma(z_f)$ and $b = \sigma(z_b)$ the three classes are fake f, beautified $(1-f)b$ and real $(1-f)(1-b)$, and the decision reads $P(\text{fake}) = f$ only. An operation head (presence and amount of four operations) and a linear layer on the 256 patch tokens, which gives a 16×16 evidence map trained against the exact footprint of each edit, are trained alongside and are not read by the decision (supplement). Each training step builds a genuine, a beautified and a forged item from one genuine FF++ c23 [44] or FFHQ frame (Fig. 2b; sources in the supplement), the forged item beautified with probability 0.5 so that both classes receive the views; the beautified renders come from two retouching services that are sister services of the test service. The test service itself is never seen. Training: 224 px, AdamW at 10^{-4}, 8 epochs; checkpoints are chosen on validation data only, by a balanced criterion that penalises originals sent to the beautified class (supplement).

Both axes. On the retouching pairs the model ranks the render above its original in 32.5 and 41.3% of pairs on its two seeds and accuses 8.6 and 12.5% (Table I); its native decision sends 98.8 and 99.2% of the renders to beautified. On Celeb-DF-B it keeps the highest AUC under the presets, 0.866 and 0.883, and its beautification-only miss rise of 6.7 and 8.3 points is smaller than that of Forensics Adapter, SBI and GenD but larger than that of DFD-HR and CADDM (Table II). CADDM is lower on both raw axes (6.3% and -2.9 points) at a beautified AUC of 0.762 (Fig. 5).

The controlled comparison is the same architecture with and without beautified views (Table III). Both-class views take the accusation from 54.2 to 11.7% and change the escape by -7.2, -4.0 and $+2.0$ points over three seeds. On the full data a forgery-head-only model is as good on both axes within seed spread (9.1 and 6.5% accused, beautified AUC 0.896 and 0.904), so the gain comes from the training data, not from the third class.

Untreated detection. On Celeb-DF-v2 and DFD the model is on par with all four CLIP detectors (frame AUC 0.891 and 0.942, video 0.947; no paired interval excludes zero in either direction; Table S3). On DF40 face swaps it is at Effort's level in the FF++ domain and 0.03 below it in the Celeb-DF domain (video AUC 0.952 and 0.905 against 0.944 and 0.935; no paired interval was computed).

Costs. Four are measured. On DF40 reenactment the model is behind: 0.718 and 0.502 in the two domains against 0.808 and 0.710 for Effort and 0.935 and 0.828 for GenD, and a

forgery-head-only model on the same data is 0.08–0.15 above it, so the shared beautified head costs reenactment (Table S33). It sends 20.8 and 19.6% of the service's untouched originals to the beautified class, a cost the retouching source pays for the low accusation. Its beautified class is itself an escape route: on Celeb-DF-B it sends 16.6 and 16.1% of the beautified deepfakes to beautified, against 3.6 and 3.9% of the same deepfakes untreated; these count as misses in Table II.

Operation scores. Its four operation scores are reliable only for smoothing: on the service's single-operation renders the presence head recovers smoothing in 94% of cases but eye enlarging in 20%, face slimming in 32% and whitening in 38%, and on real-app filters almost none (recall 0.003–0.24; Table S10). The operation names describe the training family rather than beautification in general, and we no longer claim them.

V. Limitations and What We No Longer Claim

The measurement covers one commercial retouching service, four app presets on one video dataset, seven scripted smoothing forms and one open-source beauty engine. The retouching source was read during the study and is an analysis set. GPUPixel's skin smoothing, a source no design decision touched, reproduces the accusation (14 of 15 face detectors above 15% at its strongest level, against 2.4–4.9% for the same pipeline without smoothing) but not the ranking of the detectors (Spearman 0.24 against the scripted bilateral form). It is a port of the shader, not the compiled library. Celeb-DF-B shares its genuine videos and identities with Celeb-DF-v2, on which every detector in the table was evaluated by its authors, so it is not identity disjoint. P3 and several SIC arms have one seed, and no non-CLIP pipeline was trained with the rule. Effort and DFD-HR were not reproduced at their published Celeb-DF-v2 numbers (Table S3), and we compare against their released weights.

Four claims of earlier drafts are withdrawn and one result is moved out. The FF++ texture origin of the cue: an arm trained without Deepfakes and NeuralTextures accuses as much (Sec. III-D). Instagram AR filters: the renders cannot be separated from the OpenFilter capture pipeline (Sec. II) and our models accuse them as much as the CLIP detectors (Table S9). Operation naming: the four operation scores identify smoothing on the training family and little else (Sec. IV). Language-model explanations did not differ from template sentences in any measured respect and were removed; localisation of part-level forgeries and its restoration test are in the supplement. Two further channels were tested without training and closed: weighting the texture–identity fusion by the model's own beautified estimate moves the error from misses to false alarms, since the estimate rises on beautified genuine faces as on beautified forgeries; and the pulse signal of remote photoplethysmography has no separation to lose on Celeb-DF (video AUC 0.51), because the swaps inherit the driving video's pulse (peak frequency agreement with the source video 44% against 19% by chance).

VI. Related Work

Four strands of prior work touch this measurement: beautification evaluations, benign-edit benchmarks, shortcut learning and identity evidence. Libourel et al. [3], Saeed et al. [4] and Concas et al. [5] are discussed in Sec. I; Libourel et al. close by recommending a dedicated filter-detection stage, which the companion model's beautified head is. Tariq et al. [8] benchmark detectors against moiré and name beautification as future work; von Stryk and Keuper [33] find that false alarms on smartphone photographs follow threshold calibration, which Finding 5 rules out for skin smoothing.

Retouching detection and restoration [2], [45]–[50] have no forgery class; OpenFilter [1] renders AR filters; Liu et al. [51] attribute editing tools in a closed set; perturbation benchmarks [7], [9] treat edits as corruptions of one class.

Shortcut learning [31], group shifts [52] and class-dependent augmentation [32] describe the mechanism of Finding 3; Ricker et al. [6] found the sign flip for JPEG with two cells. CORE [20] ties representations across generic augmentations and still ranks a smoothed genuine face above its original in 97% of pairs; the alpha-blending hypothesis [53] holds that frame-based detectors read compositing artefacts; our edge-treatment ablation shows that face-local skin smoothing produces that inconsistency with no boundary line at all.

Identity-referenced detection [34]–[38] and face recognition under beauty filters [39] are the prior art for Finding 7; DeepfakeBench [54] and DF40 [13] provide the checkpoints and test sets.

VII. Conclusion

Seventeen published face-forgery detectors read skin smoothing as forgery evidence: one cue accuses the smoothed genuine photograph and lets the app-beautified deepfake through. The cue is set by which class saw smoothing in training. It is the contrast between a flattened face and its untouched surroundings, which face-local beautification produces and face-swap detectors are built to flag. No threshold, routing, consistency loss or test-time rule removes it, because, once both classes are smoothed, no score separates them. What beautification does not erase is who the person is: identity against a reference keeps its separation under real app presets, provided the threshold is set on beautified genuine faces. The manifests, the rule, the paired reader, the per-detector result files and the pre-registration log are released.

References

[1] P. Riccio, B. Psomas, F. Galati, F. Escolano, T. Hofmann, and N. Oliver, "OpenFilter: A framework to democratize research access to social media AR filters," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, vol. 35, 2022, arXiv:2207.12319.

[2] Q. Ying, J. Liu, S. Li, H. Xu, Z. Qian, and X. Zhang, "RetouchingFFHQ: A large-scale dataset for fine-grained face retouching detection," in *Proceedings of the 31st ACM International Conference on Multimedia*, 2023, pp. 737–746.

[3] A. Libourel, S. Husseini, N. Mirabet-Herranz, and J.-L. Dugelay, "A case study on how beautification filters can fool deepfake detectors," in *12th International Workshop on Biometrics and Forensics (IWBF)*, 2024, pp. 1–6.

[4] M. S. Saeed, I. Ul Haq, and K. Malik, "Realism to deception: Investigating deepfake detectors against face enhancement," 2025.

[5] S. Concas, S. M. La Cava, A. Panzino, E. Masala, G. Orrù, and G. L. Marcialis, "Deceptive beauty: Evaluating the impact of beauty filters on deepfake and morphing attack detection," in *IEEE International Conference on Metrology for eXtended Reality, Artificial Intelligence and Neural Engineering (MetroXRAINE)*, 2025, pp. 61–66, arXiv:2509.14120.

[6] J. Ricker, A. Fischer, and E. Quiring, "BIAS-ID: A framework for analyzing transformation biases in AI-generated image detectors," *arXiv preprint arXiv:2605.31153*, 2026.

[7] L. Jiang, R. Li, W. Wu, C. Qian, and C. C. Loy, "DeeperForensics-1.0: A large-scale dataset for real-world face forgery detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 2889–2898.

[8] R. Tariq, M. Heo, S. S. Woo, and S. Tariq, "Through the lens: Benchmarking deepfake detectors against moiré-induced distortions," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[9] D. Hendrycks and T. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," in *International Conference on Learning Representations (ICLR)*, 2019.

[10] T. Karras, S. Laine, and T. Aila, "A style-based generator architecture for generative adversarial networks," in *CVPR*, 2019.

[11] Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu, "Celeb-DF: A large-scale challenging dataset for deepfake forensics," in *CVPR*, 2020.

[12] N. Dufour and A. Gully, "Contributing data to deepfake detection research," Google AI Blog, 2019.

[13] Z. Yan, T. Yao, S. Chen, Y. Zhao, X. Fu, J. Zhu, D. Luo, C. Wang, S. Ding, Y. Wu, and L. Yuan, "DF40: Toward next-generation deepfake detection," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, vol. 37, 2024, pp. 29 387–29 434.

[14] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.

[15] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *ICML*, 2019.

[16] Y. Qian, G. Yin, L. Sheng, Z. Chen, and J. Shao, "Thinking in frequency: Face forgery detection by mining frequency-aware clues," in *European Conference on Computer Vision (ECCV)*, 2020.

[17] H. Liu, X. Li, W. Zhou, Y. Chen, Y. He, H. Xue, W. Zhang, and N. Yu, "Spatial-phase shallow learning: Rethinking face forgery detection in frequency domain," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

[18] Z. Yan, Y. Zhang, Y. Fan, and B. Wu, "UCF: Uncovering common features for generalizable deepfake detection," in *ICCV*, 2023.

[19] J. Cao, C. Ma, T. Yao, S. Chen, S. Ding, and X. Yang, "End-to-end reconstruction-classification learning for face forgery detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[20] Y. Ni, D. Meng, C. Yu, C. Quan, D. Ren, and Y. Zhao, "CORE: COnsistent REpresentation learning for face forgery detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2022.

[21] Y. Luo, Y. Zhang, J. Yan, and W. Liu, "Generalizing face forgery detection with high-frequency features," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

[22] K. Shiohara and T. Yamasaki, "Detecting deepfakes with self-blended images," in *CVPR*, 2022.

[23] S. Dong, J. Wang, R. Ji, J. Liang, H. Fan, and Z. Ge, "Implicit identity driven deepfake face swapping detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 4490–4499.

[24] D. Nguyen, N. Mejri, I. P. Singh, P. Kuleshova, M. Astrid, A. Kacem, E. Ghorbel, and D. Aouada, "LAA-Net: Localized artifact attention network for quality-agnostic and generalizable deepfake detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 17 395–17 405.

[25] Z. Yan, J. Wang, P. Jin, K.-Y. Zhang, C. Liu, S. Chen, T. Yao, S. Ding, B. Wu, and L. Yuan, "Orthogonal subspace decomposition for generalizable AI-generated image detection," in *ICML*, ser. Proceedings of Machine Learning Research, vol. 267, 2025, pp. 70 268–70 288.

[26] X. Cui, Y. Li, A. Luo, J. Zhou, and J. Dong, "Forensics adapter: Adapting CLIP for generalizable face forgery detection," in *CVPR*, 2025, pp. 19 207–19 217.

[27] A. Yermakov, J. Cech, J. Matas, and M. Fritz, "Deepfake detection that generalizes across benchmarks," in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026.

[28] J. Sun, Z. Yan, K.-Y. Zhang, T. Yao, and S. Ding, "DFD-HR: Generalizable deepfake detection via hierarchical routing learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026, pp. 13 984–13 995.

[29] U. Ojha, Y. Li, and Y. J. Lee, "Towards universal fake image detectors that generalize across generative models," in *CVPR*, 2023.

[30] C. Tan, H. Liu, Y. Zhao, S. Wei, G. Gu, P. Liu, and Y. Wei, "Rethinking the up-sampling operations in CNN-based generative network for generalizable deepfake detection," in *CVPR*, 2024, pp. 28 130–28 139.

[31] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut learning in deep neural networks," *Nature Machine Intelligence*, vol. 2, no. 11, pp. 665–673, 2020.

[32] R. Balestriero, L. Bottou, and Y. LeCun, "The effects of regularization and data augmentation are class dependent," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, 2022, pp. 37 878–37 891.

[33] A. von Stryk and J. Keuper, "LAION-Mobile: Evaluating deepfake detectors on one million smartphone photos," 2026.

[34] X. Dong, J. Bao, D. Chen, T. Zhang, W. Zhang, N. Yu, D. Chen, F. Wen, and B. Guo, "Protecting celebrities from deepfake with identity consistency transformer," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 9468–9478.

[35] X. Dong, J. Bao, D. Chen, W. Zhang, N. Yu, D. Chen, F. Wen, and B. Guo, "Identity-driven deepfake detection," *arXiv preprint arXiv:2012.03930*, 2020.

[36] D. Cozzolino, A. Rössler, J. Thies, M. Nießner, and L. Verdoliva, "ID-Reveal: Identity-aware deepfake video detection," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, arXiv:2012.02512.

[37] Z. Yin and R. A. Jamaluddin, "Identity-referenced two-stage deepfake detection with consistency-gated fusion," *Journal of Information System and Technology Management*, vol. 11, no. 44, p. 904, 2026.

[38] K. Lin, Z. Yan, K.-Y. Zhang, L. Hao, Y. Zhou, Y. Lin, W. Li, T. Yao, S. Ding, and B. Li, "Guard me if you know me: Protecting specific face-identity from deepfakes," *arXiv preprint arXiv:2505.19582*, 2025.

[39] N. Mirabet-Herranz, C. Galdi, and J.-L. Dugelay, "Facial biometrics in the social media era: An in-depth analysis of the challenge posed by beautification filters," *IEEE Transactions on Biometrics, Behavior, and Identity Science*, vol. 7, no. 1, 2025.

[40] C. Rathgeb, C.-I. Satnoianu, N. E. Haryanto, K. Bernardo, and C. Busch, "Differential detection of facial retouching: A multi-biometric approach," *IEEE Access*, vol. 8, pp. 106 373–106 385, 2020.

[41] J. Guo, X. Zhu, Y. Yang, F. Yang, Z. Lei, and S. Z. Li, "Towards fast, accurate and stable 3D dense face alignment," in *European Conference on Computer Vision (ECCV)*, 2020, pp. 152–168.

[42] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *ICML*, 2021.

[43] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *ICLR*, 2022.

[44] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, "FaceForensics++: Learning to detect manipulated facial images," in *ICCV*, 2019.

[45] A. Bharati, R. Singh, M. Vatsa, and K. W. Bowyer, "Detecting facial retouching using supervised deep learning," *IEEE Transactions on Information Forensics and Security*, vol. 11, no. 9, pp. 1903–1913, 2016.

[46] A. Jain, R. Singh, and M. Vatsa, "On detecting GANs and retouching based synthetic alterations," in *IEEE International Conference on Biometrics Theory, Applications and Systems (BTAS)*, 2018, pp. 1–7.

[47] S.-Y. Wang, O. Wang, A. Owens, R. Zhang, and A. A. Efros, "Detecting photoshopped faces by scripting photoshop," in *ICCV*, 2019.

[48] F. Xing, X. Shi, Y.-G. Wang, and C. Yang, "Unveiling authenticity with diffusion-based face retouching reversal," *Applied Soft Computing*, vol. 174, 2025.

[49] J. Liu, Q. Ying, Z. Qian, S. Li, R. Zhang, J. Liu, and X. Zhang, "MoFRR: Mixture of diffusion models for face retouching restoration," in *ICCV*, 2025, pp. 12 842–12 851.

[50] G. Zhao, Y. Gu, X. Sheng, Y. Hu, and J. Zhang, "Label-guided facial retouching reversion," in *ICME*, 2025.

[51] S. Liu, Q. Sheng, D. Wang, Y. Li, C. Zhou, and J. Cao, "Multi-tool image editing attribution in facial forgery," 2026.
[52] S. Sagawa, P. W. Koh, T. B. Hashimoto, and P. Liang, "Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization," in *International Conference on Learning Representations (ICLR)*, 2020.
[53] A. Yermakov, J. Cech, M. Fritz, and J. Matas, "The alpha blending hypothesis: Compositing shortcut in deepfake detection," 2026.
[54] Z. Yan, Y. Zhang, X. Yuan, S. Lyu, and B. Wu, "DeepfakeBench: A comprehensive benchmark of deepfake detection," in *NeurIPS Datasets and Benchmarks Track*, 2023.